

Employability Skills Development

Course Code:	2418408 (T2400408)
Programme:	Diploma in Computer Science & Information Technology
Semester:	IV
Board:	SBTE, Bihar
Credits:	01 Total Marks: 25

Course Outcomes (COs)

CO-1: Build resume and showcase portfolio for placement activity.

CO-2: Face interviews and participate effectively in Group Discussions.

CO-3: Apply engineering tools in work situations and societal processes.

Rationale

Education may only be enough to qualify for a job, but **employability skills** are the major criteria considered for a job role. Employability skills are the building blocks of any career — they equip one to carry out roles in the company to the best of their ability. Employers check these skills before hiring. These job-readiness skills are behaviors and essential attitudes that enable students to grow in their careers. Employers value them because they indicate how well an employee will get along with team members and customers, and how efficiently they will handle job performance and career success.

UNIT 1 — GOAL SETTING

Relevant CO: CO-1 (Build resume and showcase portfolio for placement activity)

1.1 Career Planning and SWOT Analysis

Career planning is the process of identifying your professional goals and creating a path to achieve them. It involves self-assessment, exploration of options, and setting actionable steps.

SWOT Analysis

SWOT stands for **Strengths, Weaknesses, Opportunities, and Threats**. It is a self-assessment tool used to understand one's current position and plan future growth.

SWOT	Definition	Examples
Strengths (S)	Internal positive attributes you control	Technical skills, communication, certifications

Weaknesses (W)	Internal factors that may hinder progress	Lack of experience, poor time management
Opportunities (O)	External factors you can exploit	Job openings, new technologies, internships
Threats (T)	External factors that may challenge you	Competition, automation, economic downturn

Steps in Career Planning:

1. **Self-Assessment** — Identify your interests, values, skills, and personality traits.
2. **Career Exploration** — Research various careers, industries, and job roles.
3. **Goal Setting** — Set SMART goals (Specific, Measurable, Achievable, Relevant, Time-bound).
4. **Action Plan** — List the steps, resources, and timeline needed to achieve goals.
5. **Review and Revise** — Periodically evaluate progress and adjust plans accordingly.

1.2 Resume Building Using Internet Formats

A **resume** (also called a CV — Curriculum Vitae) is a formal document summarizing your education, work experience, skills, and achievements. It is the first impression you make on an employer.

Key Sections of a Resume:

- **Personal Information:** Full name, address, phone number, email, LinkedIn/GitHub links.
- **Career Objective:** A brief 2–3 line statement describing your career goals and what you offer.
- **Educational Qualifications:** Degrees, institutions, year of passing, percentage/CGPA.
- **Technical Skills:** Programming languages, tools, software, hardware expertise.
- **Projects / Internships:** Title, duration, technologies used, your role, and outcomes.
- **Certifications / Courses:** Online certifications (Coursera, NPTEL, etc.) relevant to your field.
- **Extra-Curricular Activities:** Participation in fests, clubs, competitions, volunteering.
- **Declaration:** A statement confirming the accuracy of information provided.

Types of Resume Formats:

- **Chronological:** Lists work experience in reverse chronological order. Best for experienced candidates.
- **Functional:** Focuses on skills rather than job history. Suitable for freshers or career changers.
- **Combination:** Merges both skills and chronological experience. Most versatile format.

Online Resume Building Tools:

- Canva (canva.com)
- Novoresume (novoresume.com)
- Resume.io
- Zety
- LinkedIn Resume Builder

Do's and Don'ts of Resume Writing:

DO's	DON'Ts
Keep it concise (1–2 pages)	Add unnecessary personal details (religion, caste)
Use action verbs (Developed, Implemented, Designed)	Use vague language ('helped with tasks')
Tailor resume for each job application	Include false information

Use a professional email address	Use decorative fonts or excessive colors
Proofread for grammar and spelling	Leave unexplained gaps in timeline

1.3 Showcase Portfolio

A **portfolio** is a collection of work samples, projects, achievements, and evidence of your skills and capabilities. Unlike a resume (which tells), a portfolio *shows* what you can do.

What to Include in a Portfolio:

- Academic projects with code/documentation links (GitHub)
- Certifications and course completion letters
- Photographs/videos of hardware projects or lab work
- Research papers, reports, or technical write-ups
- Awards, appreciation letters, competition results
- Internship work samples (with employer permission)
- Self-learning evidence: MOOCs, spoken tutorials, etc.

Types of Portfolios:

- **Physical Portfolio:** A printed binder or folder with documents, certificates, and project prints.
- **Digital Portfolio:** A PDF or online webpage showcasing work. Easy to share via email/links.
- **Website Portfolio:** A personal website (using WordPress, GitHub Pages, Wix) displaying all work professionally.

1.4 Personal Grooming

Personal grooming refers to the practice of maintaining cleanliness, tidiness, and a professional appearance. First impressions in interviews are formed within seconds — appearance plays a key role.

Key Aspects of Personal Grooming:

- **Dress Code:** Wear clean, ironed, and professionally appropriate clothing. Formal attire for interviews.
- **Hair:** Keep hair clean, well-combed, and neatly styled.
- **Hygiene:** Maintain dental hygiene, use deodorant, keep nails trimmed.
- **Posture:** Stand and sit straight; avoid slouching.
- **Eye Contact:** Maintain steady eye contact to convey confidence.
- **Accessories:** Keep accessories minimal and professional.

1.5 Self-Introduction

Self-introduction is the first opportunity to make an impression in interviews, group discussions, and professional settings. A well-structured introduction is concise, confident, and relevant.

Structure of a Good Self-Introduction:

1. **Greeting:** 'Good morning/afternoon, sir/ma'am.'
2. **Name:** 'My name is [Full Name].'
3. **Education:** 'I am pursuing Diploma in CSE from [College], Semester IV.'
4. **Hometown:** 'I am from [City], Bihar.'
5. **Skills:** 'My key technical skills include Java, DBMS, and Python.'
6. **Strengths:** 'I am a quick learner, hardworking, and team player.'

7. **Goals:** 'I aspire to work as a software developer and contribute to meaningful projects.'

8. **Closing:** 'Thank you for this opportunity.'

Tips for Effective Self-Introduction:

- Keep it between 60–90 seconds
- Speak clearly and at a moderate pace
- Avoid memorizing word-for-word; speak naturally
- Do not read from paper
- Practice in front of a mirror or record yourself

1.6 Website Development (Personal Website)

A **personal website** serves as an online portfolio and professional presence. It showcases your identity, skills, projects, and contact information to potential employers.

Sections of a Personal Website:

- Home / About Me
- Skills
- Projects
- Education
- Certifications
- Contact / Social Links

Popular Tools/Platforms to Build Personal Website:

- **GitHub Pages:** Free, uses HTML/CSS/JS. Ideal for developers. Domain: username.github.io
- **WordPress:** Easy drag-and-drop. Requires hosting or free wordpress.com plan.
- **Wix / Weebly:** No coding needed. Drag-and-drop interface. Free plan available.
- **Google Sites:** Very simple tool by Google. Suitable for basic portfolios.

UNIT 2 — CAPACITY DEVELOPMENT

Relevant CO: CO-2 (Face interviews and participate effectively in Group Discussions)

2.1 Interview Skills

An **interview** is a formal conversation between an employer and a candidate to assess suitability for a job. Interview skills are the techniques that help you present yourself effectively.

Types of Interviews:

- **Face-to-Face Interview:** Traditional one-on-one or panel interview in person.
- **Telephonic Interview:** Preliminary screening conducted over phone.
- **Video/E-Interview:** Conducted via Zoom, Google Meet, Skype, or MS Teams.
- **Group Interview:** Multiple candidates interviewed together; assesses teamwork and leadership.
- **Technical Interview:** Tests domain-specific knowledge (coding, problem-solving).
- **HR Interview:** Assesses personality, communication, salary expectations, cultural fit.
- **Walk-in Interview:** No prior appointment needed; candidates arrive directly.

Stages of an Interview Process:

1. Pre-interview Preparation (research company, prepare answers, dress appropriately)
2. Introduction and Ice-breaking
3. Technical / Domain Questions
4. Behavioral Questions (STAR method: Situation, Task, Action, Result)
5. HR Round (salary, joining date, goals)
6. Candidate Questions to Interviewer
7. Closing and Thank-you

Common Interview Questions and How to Answer:

- **Q: Tell me about yourself**

Tip: Use the Self-Introduction structure from Unit 1.

- **Q: What are your strengths and weaknesses?**

Tip: State genuine strengths; show weakness as an area you are improving.

- **Q: Why do you want this job?**

Tip: Align your skills and goals with the company's mission.

- **Q: Where do you see yourself in 5 years?**

Tip: Show ambition and long-term commitment to growth.

- **Q: Why should we hire you?**

Tip: Highlight unique skills, relevant projects, and problem-solving ability.

E-Interview Tips (Online Interview):

- Test your camera, microphone, and internet connection before the interview
- Choose a clean, well-lit, quiet background
- Dress professionally (full formal, not just top)
- Look at the camera, not at your own image
- Keep your phone on silent and notifications off
- Have a backup plan (mobile hotspot) in case of connectivity issues
- Keep your resume/notes nearby but do not read from them

2.2 Group Discussion (GD)

A **Group Discussion** is a structured discussion among a group (usually 6–12 people) on a given topic. It is used by employers to evaluate communication, leadership, teamwork, and critical thinking.

Purpose of GD:

- Test communication and listening skills
- Assess leadership and team participation
- Evaluate ability to think on the spot
- Check knowledge and analytical ability
- Observe confidence and assertiveness

Do's and Don'ts of Group Discussion:

DO's in GD	DON'Ts in GD
Listen actively to others	Interrupt rudely or shout
Speak clearly and confidently	Monopolize the discussion
Support your points with examples/data	Go off-topic or repeat the same point

Initiate the GD if possible (shows leadership)	Use informal or slang language
Invite quieter members to speak	Make personal attacks on others' views
Summarize the discussion at the end	Change your opinion without a logical reason
Use transitions: 'I agree with...', 'Adding to that...'	Remain silent throughout

Roles in a Group Discussion:

- **Initiator:** Starts the discussion; shows confidence and knowledge.
- **Contributor:** Adds relevant points at appropriate times.
- **Moderator:** Keeps discussion on track; manages time.
- **Summarizer:** Concludes the discussion with a concise summary.

How to Interrupt Politely:

- 'Excuse me, I'd like to add a point here...'
- 'That is a valid point, however I believe...'
- 'With due respect, I would like to present a different perspective...'
- 'Building on what [name] said, I think...'

Synthesis and Analysis in GD:

Analysis means breaking down the topic into its components — identifying causes, effects, pros and cons, and different viewpoints. **Synthesis** means combining various points from the discussion into a coherent conclusion that captures the group's overall understanding.

2.3 Social Media for Personal Enrichment and Netiquette

Netiquette (Internet Etiquette) refers to the code of conduct for respectful and professional behavior on the internet and social media platforms.

Professional Use of Social Media:

- **LinkedIn:** Build professional network, post achievements, connect with employers, follow industry news.
- **GitHub:** Showcase code projects; contribute to open source; employer profile visibility.
- **Twitter/X:** Follow tech leaders, share knowledge, participate in tech conversations.
- **YouTube:** Watch tutorials, follow educational channels (NPTEL, freeCodeCamp, CS50).
- **Instagram:** Can be used for personal branding if managed professionally.

Netiquette Rules:

- Think before you post — once online, content may be permanent
- Do not share fake news or unverified information
- Respect others' opinions; avoid cyberbullying
- Use appropriate language — avoid slang or offensive terms in professional spaces
- Protect privacy — do not share others' personal information
- Give credit when sharing others' work (attribution)
- Respond to messages and comments professionally and promptly

2.4 Body Language

Body language is non-verbal communication through posture, gestures, facial expressions, and eye contact. Research shows up to **55% of communication is non-verbal**.

Key Elements of Positive Body Language:

- **Eye Contact:** Maintains trust and confidence. Look at the interviewer/audience, not the floor.
- **Posture:** Sit/stand straight with shoulders back. Avoid slouching or crossing arms.
- **Handshake:** Firm (not crushing), brief, and confident. Used in formal greetings.
- **Facial Expressions:** Smile naturally; show genuine interest and enthusiasm.
- **Gestures:** Use open hand gestures to emphasize points; avoid pointing fingers.
- **Head Nodding:** Shows active listening and agreement.
- **Foot Position:** Keep feet flat on the floor; avoid nervous tapping.
- **Personal Space:** Maintain appropriate distance (typically 1–1.5 meters in professional settings).

Negative Body Language to Avoid:

- Avoiding eye contact (signals dishonesty or nervousness)
- Crossing arms (signals defensiveness)
- Fidgeting (signals nervousness or boredom)
- Pointing or aggressive gestures
- Looking at watch or phone (signals disinterest)

2.5 Self-Management

Self-management is the ability to regulate one's own behavior, emotions, and time effectively. It is a critical soft skill for career success.

Components of Self-Management:

- **Time Management:** Prioritize tasks using tools like to-do lists, calendars. Follow the Eisenhower Matrix (Urgent vs Important).
- **Goal Setting:** Set SMART goals and break them into actionable daily/weekly tasks.
- **Self-Discipline:** Follow through on commitments; resist distractions (especially social media).
- **Stress Management:** Use techniques like deep breathing, exercise, meditation, and adequate sleep.
- **Continuous Learning:** Develop a habit of learning new skills through MOOCs, books, and practice.
- **Accountability:** Take ownership of your mistakes; learn from failures rather than blaming others.

2.6 Emotional Intelligence (EI / EQ)

Emotional Intelligence is the ability to recognize, understand, manage, and effectively use emotions — both your own and others'. Popularized by psychologist **Daniel Goleman**.

Five Components of Emotional Intelligence (Goleman's Model):

- **1. Self-Awareness:** Knowing your own emotions, strengths, weaknesses, and how they affect others.
- **2. Self-Regulation:** Controlling or redirecting disruptive emotions. Thinking before acting.
- **3. Motivation:** Being driven by internal factors; having a passion for your work beyond money/status.
- **4. Empathy:** Understanding others' emotions and perspectives; essential for teamwork and leadership.
- **5. Social Skills:** Managing relationships, building networks, finding common ground, leading change.

Why EI Matters in the Workplace:

- High EI employees handle pressure and conflict better

- They build stronger professional relationships
- They are better leaders and team players
- EI predicts job performance more accurately than IQ in many roles
- Organizations prefer employees who can collaborate and communicate effectively

How to Improve EI:

- Practice self-reflection — journal your thoughts and reactions
- Seek feedback from peers and mentors
- Practice active listening in conversations
- Develop empathy by considering others' perspectives
- Manage stress through mindfulness and relaxation techniques

UNIT 3 — UTILIZING POTENTIAL

Relevant CO: CO-3 (Apply engineering tools in work situations and societal processes)

3.1 Social Networking

Social networking in a professional context means building and maintaining relationships with people who can help you in your career — mentors, peers, industry professionals, and alumni.

Types of Professional Networking:

- **Online Networking:** LinkedIn connections, GitHub collaborations, online forums (Stack Overflow).
- **Offline Networking:** College fests, industry events, job fairs, seminars, alumni meets.
- **Peer Networking:** Building relationships with classmates and batch-mates.
- **Mentor Networking:** Connecting with seniors, professors, or industry professionals for guidance.

Tips for Effective Networking:

- Have a clear, updated LinkedIn profile with a professional photo
- Reach out with personalized messages, not generic requests
- Attend technical events, hackathons, and workshops
- Offer value before asking for help (share knowledge, help others)
- Follow up after meeting someone — send a thank-you message
- Join technical communities: GDSC, ACM, IEEE student chapters

3.2 Social Engagement and Volunteering

Social engagement means actively participating in activities that contribute to the welfare of your community. Volunteering builds empathy, leadership, and teamwork — skills valued by employers.

Benefits of Volunteering:

- Develops leadership and project management skills
- Enhances resume with practical experience
- Builds a sense of civic responsibility
- Expands professional network
- Improves communication and interpersonal skills

Examples of Social Engagement for Diploma Students:

- Teaching basic computer skills to underprivileged children
- Participating in Swachh Bharat or tree plantation drives
- Organizing digital literacy camps in villages
- Contributing to open-source projects that benefit society
- Participating in NSS (National Service Scheme) activities

3.3 Collaboration and Teamwork

Teamwork is the combined effort of a group to achieve a shared goal. Modern engineering workplaces require professionals who can collaborate effectively across teams and disciplines.

Characteristics of Effective Teams:

- Clear and shared goals
- Defined roles and responsibilities
- Open and honest communication
- Mutual respect and trust
- Constructive handling of conflict
- Accountability — each member owns their tasks
- Celebrating team successes together

Stages of Team Development (Tuckman's Model):

- **Forming:** Team members meet, roles are unclear, polite behavior.
- **Storming:** Conflicts arise as personalities and ideas clash.
- **Norming:** Team establishes rules, roles become clear, cooperation begins.
- **Performing:** Team works efficiently toward goals with high productivity.
- **Adjourning:** Project ends; team reflects on achievements and disbands.

Tips for Effective Collaboration:

- Use project management tools: Trello, Jira, GitHub Projects
- Hold regular check-in meetings; use agendas
- Share workload fairly; play to each member's strengths
- Give constructive feedback — focus on the work, not the person
- Document decisions and action items after meetings

3.4 QC Tools (Quality Control Tools)

QC Tools are analytical techniques used to identify, analyze, and solve quality problems in engineering and manufacturing environments. Also known as the **7 Basic Quality Tools**.

1. Check Sheet

Definition: A structured form for collecting data in real-time at the location where data is generated.

Example: Tracking frequency of defects in software or hardware testing.

2. Fishbone Diagram (Cause & Effect / Ishikawa Diagram)

Definition: A visual tool that identifies potential causes of a problem. Shaped like a fish skeleton.

Example: Finding root causes of a software bug or machine failure.

3. Histogram

Definition: A bar chart showing the frequency distribution of data.

Example: Analyzing the distribution of response times in a network.

4. Pareto Chart

Definition: A bar chart combined with a line graph. Based on the 80/20 rule — 80% of problems arise from 20% of causes.

Example: Identifying the most frequent types of errors in code.

5. Control Chart (Shewhart Chart)

Definition: A graph with control limits used to monitor process stability over time.

Example: Monitoring server uptime or CPU usage over time.

6. Scatter Diagram

Definition: Shows the relationship (correlation) between two variables.

Example: Relationship between code complexity and number of bugs.

7. Stratification (Flow Chart)

Definition: Separating data into subgroups to identify patterns or differences.

Example: Comparing defects across different software modules.

3.5 Lean Manufacturing Techniques

Lean manufacturing is a methodology that focuses on minimizing waste within manufacturing systems while simultaneously maximizing productivity. Originated from the Toyota Production System (TPS).

Key Lean Concepts:

Kanban

Concept: A visual workflow management method. Tasks are represented as cards on a board with columns (To Do, In Progress, Done). Limits work-in-progress to prevent bottlenecks.

IT Application: Used in software development: GitHub Projects, Trello boards.

Kaizen

Concept: Japanese word meaning 'change for the better' or 'continuous improvement'. Small, incremental improvements made regularly by all employees at all levels.

IT Application: Daily stand-up meetings to discuss yesterday's work, today's plan, and blockers (Agile Scrum).

Five S (5S)

Concept: A workplace organization method for efficiency and safety.

IT Application: Organizing a computer lab or software team workspace.

Poka-Yoke

Concept: Meaning 'mistake-proofing' in Japanese. Designing processes so that errors are prevented or detected immediately.

IT Application: Form validation in web apps — prevents incorrect data entry before submission.

Quality Circle

Concept: A small group of employees who do similar work and regularly meet to identify, analyze, and solve work-related quality problems.

IT Application: A team of programmers meeting weekly to review code quality and suggest improvements.

The 5S Framework (Detailed):

- **Sort (Seiri):** Remove unnecessary items from the workspace. Keep only what is needed.
- **Set in Order (Seiton):** Organize remaining items so they are easy to find and use. 'A place for everything.'
- **Shine (Seiso):** Clean the workspace regularly. Identify and eliminate sources of dirt/clutter.
- **Standardize (Seiketsu):** Establish standards/procedures to maintain the first 3 S's consistently.
- **Sustain (Shitsuke):** Make 5S a habit. Regular audits; discipline and commitment to maintain the system.

5S in Software/IT Context:

- Sort: Delete unused files, old versions, deprecated code
- Set in Order: Organize code into proper folder structures; use naming conventions
- Shine: Regular code reviews; clean up dead code and outdated documentation
- Standardize: Use consistent coding standards (PEP 8, Google Style Guide)
- Sustain: Conduct regular code audits; enforce standards via linters and CI/CD pipelines

Quick Revision Summary

Unit	Title	Key Topics	CO
1	Goal Setting	SWOT Analysis, Career Planning, Resume Building, Portfolio, Grooming, Self-Introduction, Personal Website	CO-1
2	Capacity Development	Interview Skills, Group Discussion, Social Media & Netiquette, Body Language, Self-Management, Emotional Intelligence	CO-2
3	Utilizing Potential	Social Networking, Volunteering, Teamwork, QC Tools (7 Basic), Lean Tools: Kanban, Kaizen, 5S, Poka-Yoke, Quality Circle	CO-3